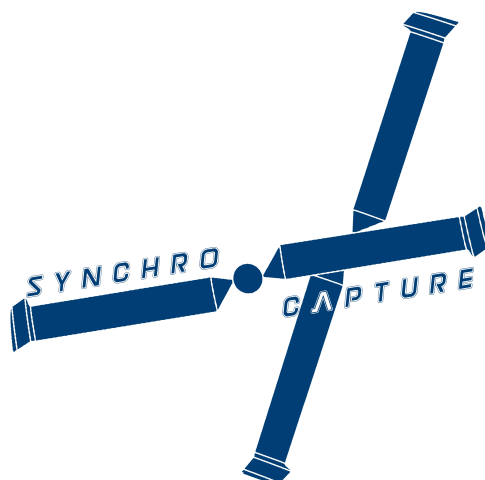


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Weight Sizing, Subsystems & HUMS Design of a Novel Synchropter Concept for the Aerial-Recovery of Falling Rocket Stages



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Abstract

An in-house methodology to estimate the component weights of a rotorcraft has been developed to size the preliminary weights of a Novel Synchropter vehicle. Within the sizing loops, the vehicle configuration defined which components were necessary and which were not. Thus, the preliminary maximum take-off weight estimation was found to be **2907 kg**. The report also outlines the fuel system and Health and Usage Monitoring System (HUMS) design. Additionally, the combustion systems design has been presented.

Acknowledgements

The author would like to acknowledge the help received from all supervisors during the duration of the project. The team also needs to be acknowledged for their outstanding technical contributions, emotional supports, and the amazing remote group environment.

"Helicopters really can't fly - they're just so ugly that the Earth repels them."

Anonymous

WRR Team Members

Table 1: *Team Members and Roles.*

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Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
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†Sub-team leads are shown in **bold**

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Nomenclature

A	Area	$[ft^2]$
AR	Diffuser Area Ratio	-
d	Diameter	[mm]
H	Radial Height	[inches]
L	Length	[inches]
LL	Lay Length	[mm]
P	Power	[hp]
P	Pressure	[kPa]
r	Radius	[inches]
sfc	Specific Fuel Consumption	[lb/h.p. hr]
t	Time Taken	[hours]
T	Temperature	[K or $^{\circ}R$]
W	Weight	[kg]
ϵ	Strain	-
α	Helix Pitch Angle	[radians]
ρ	Radius of Curvature	[mm]
μ	Air Mass Fraction	-

Abbreviations

BEMT	Blade Element Momentum Theory
CG	Centre of Gravity
EFI	Electronic Fuel Injection
FBG	Fibre Bragg Gratings
HUMS	Health and Usage Monitoring System
MTOW	Maximum Take-Off Weight
RPM	Revolution per Minute
VTOL	Vertical Take-Off Landing Aircraft

Subscripts

c	Cooling
d	Dump Diffuser
dz	Dilution Zone
f	Fibre Helix
g	Liner Gas
i	Inner Casing
m	Flat-wall Diffuser
o	Outer Casing
pz	Primary Zone
r	Rope
sz	Secondary Zone

Superscripts

*	Choked Conditions
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Introduction

This third year Group Design Project (GDP) aims to develop a preliminary design for the mid-air recovery aspect of a wholly reusable launch system for which as much as possible can be recovered and reused for subsequent launches of LEO satellites. This report forms a subset of the marking criteria for the *AERO 96005 - Group Design Project* module, seeking to test engineering judgement and design trade offs within group settings [1].

Part I

Pre-Conceptual

The first week of the project focused on the concept ideation and aerial vehicle selection of the broader category of recovery procedure. Investigation and evaluation of a VTOL as the recovery vehicle and different recovery mechanisms were performed. Three different configurations of a novel VTOL aircraft were explored. This section provides an overview of the work done on basic sizing, analysis of the ability to catch multiple payloads, and estimation of power requirement of a tiltrotor concept.

Power Estimation

The power requirement for the tiltrotor aircraft UAV configuration was estimated subsequent to estimation of empty weight, by analysing the required power in the most constraining case, in this case hovering at 20,000ft with payload, using actuator disc theory and accounting for power loss with increasing altitude. Total dry weight of engines from historical VTOL data were used to estimate the engine dry weight of this configuration using linear regression to payload mass, and the power to weight ratio was obtained. Once the power requirements were calculated, fuel weights required for hover and cruise were estimated.

Capture Mechanism

A "Multiple Hooks" concept was proposed, although discarded in the early stages of the project, as an upgraded version of the "Hook Parafoil" concept. This aspect of the project is explained in the related Capture Architecture Design report [2] in depth.

This concept suggests multiple hooks on the slung line for multiple stage capture. The hooks connecting to a winch reels the first capture in towards the vehicle to reduce moment and unnecessary forces generated, before lowering the subsequent hook and prepare for second capture. The reeled in stage is attached to where the fuselage would conventionally be before second capture, easing the capture process and resulting in one stage stored in fuselage space and another hanging below the vehicle during return.

The probability of success was assumed to reduce in an exponential order as number of stages needed to be recovered increases, thus given a Technological Readiness Level (TRL) score of 6. Figure 1 below shows the recovery procedure of the concept.



Figure 1: A "Multiple Hooks" concept.

Final Decision

A helicopter was proved to be the best option compared to the VTOL and Aircraft concept after several trade off and parameter weighting studies. The following sections of this report presents the preliminary weight sizing, fuel system and HUMS design, and the combustion system sizing of a novel helicopter concept.

Part II

Preliminary Design

1 Preliminary Weight Sizing

Most of the work presented in this section was done in collaboration with Janusz, Z.. It is advised to read the related Weight Sizing and Subsystem Design report [3] prior to this section as it details work done and decisions made before arriving to the final choice of methodology. Although redundancies with [3] have been avoided, there is some overlap to maintain the coherence of the reports on their own.

A sizing code was developed in-house using the methodology of Tischenko [4] for rotorcraft sizing. The US Army's AFDD (Aero Flight Dynamics Directorate) empty weight model [5] was used to determine preliminary estimates of most component empty weights, and modified momentum theory was used to model aerodynamics in hover, forward flight, climb, and descent to calculate the required power. The basic AFDD model was found to be sufficient for a first-order estimate of component weights, with exact weights being determined later in more detailed design. This section describes the specifics of the sizing algorithm.

1.1 Design Drivers

The sizing code uses an iterative process that begins with the specification of inputs. First, the constraints for sizing the vehicle were defined. The fixed design variables specified is shown below, and were obtained from the aerodynamics team, refer to the related Rotor Design report [6]:

- Disc Loading
- Blade aspect ratio
- Hover tip speed
- Propulsive efficiency
- Rotational frequency
- Number of blades

1.2 Sizing Methodology

Component weights such as a tail rotor, and 60% of the fuselage, were eliminated from the sizing algorithm. Subsystems weights were also modified to reflect Syn-

chroCapture's configuration. An overview of the iterative methodology is shown in Figure 2.

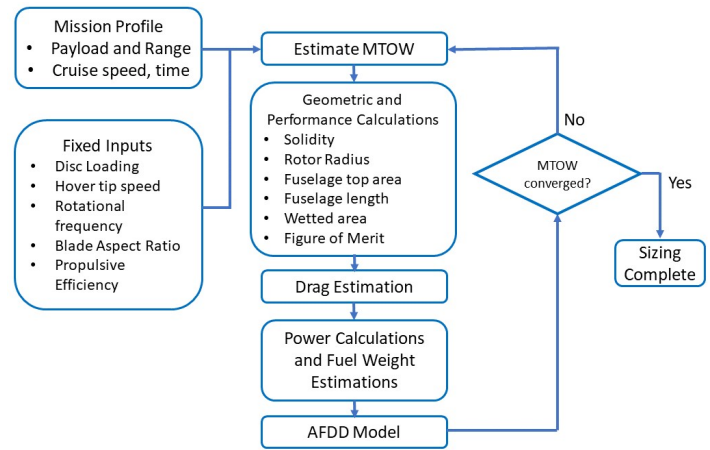


Figure 2: Sizing methodology layout.

The methodology requires a set of user-defined inputs which describe the vehicle geometry and the mission which the vehicle is expected to perform. All other parameters, including rotor radius, solidity, tail parameters and dimensions are then calculated within the code by the aerodynamics team, detailed in [6] and in the related Aerodynamic Optimisation report [7]. The sizing mission was specified in terms of its hover, climb, and cruise segments.

Within the sizing loop, geometric and drag calculations are first performed. An initial MTOW estimation of 3600 kg was made [3] as theoretically any value will converge to the same final MTOW. Using this information, the aerodynamics team, detailed in [6] and in the related Fuselage Aerodynamics report [8], provide values including:

Geometric parameters: Performance parameters:

- | | |
|----------------------|----------------------------------|
| 1. Solidity | 1. Figure of merit |
| 2. Rotor radius | 2. Drag coefficient |
| 3. Fuselage top area | 3. Vertical drag to weight ratio |
| 4. Fuselage length | |
| 5. Wetted area | |

Once the power required and fuel weights are calculated (detailed in section 1.3), the individual component masses can then be calculated using the AFDD method. It should be noted that all technology factors were set to be 1.0 [5]. Engine weight was sized using

historical data of turboshafts [9]. The take-off weight is calculated as the sum of the component weights, total fuel weight and payload weight. Finally, the take-off weight at the end of the iteration is compared with the MTOW at the start of the iteration. If the difference between the two values is below the specified tolerance of 0.5%, the sizing is considered converged. If not, the process repeats, replacing the previous initial take-off weight with the value computed by the previous iteration until the MTOW converges. This gave an output of **2907 kg**, which had a difference of 0.24% compared to the previous iteration of 2900 kg. This value was frozen because a 7 kg difference was insufficient to provide observable change in the parameters provided in the sizing loop. Figure 3 presents the take-off weight of each iteration.

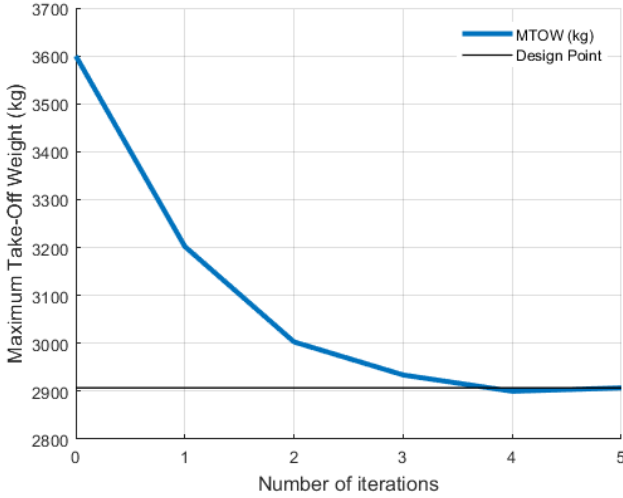


Figure 3: Iteration history of take-off weight.

The final output of the sizing algorithm was used as a starting point for completing detailed component analysis of the all component and system designs. As component designs became more detailed, the sizing algorithm was continuously updated to more accurately represent the vehicle's component weights.

1.3 Power & Fuel Weights Sizing

Geometric and performance parameters shown above are needed to calculate power and fuel required to complete the sizing mission. Power was calculated based on modified momentum theory method, considering inefficiencies due to profile and induced losses (relative

to sea level (SL), the power available at altitude is $\frac{P_{alt}}{P_{sl}} = (\frac{\rho_{alt}}{\rho_{sl}})^{0.7}$, and taking into account drag contributions during hover and forward flight [8]. Calculations of induced power also accounted for overlap area between the rotors. A more detailed explanation of engine weight sizing and power calculations for each mission segment can be found in the related Turboshaft Design report [9].

The sizing mission was discretised into different mission segments. Two values of power were calculated for each mission segment; the maximum power, calculated with full fuel load at the beginning of the segment, and the mid-weight power, calculated with half the fuel required for the particular segment consumed. The mission segment that requires most maximum power was used to size the installed power of the engine, and the most constraining case was decided to be hovering at 12,000ft capturing payload, found to be 1600 hp. It was also assumed that the average power for each mission segment equals to the mid-weight power. Fuel for each segment was then calculated for the engine running at mid-weight power constantly for the time taken to complete the mission segment. This was done by iterating Equation 1, where the mid-weight power, P_{mw} , is a function of the fuel for segment itself, until the difference between W_{fuel} on the left and right hand side of the equation converges below a set tolerance of 1%.

$$W_{fuel} = P_{mw} \cdot sfc \cdot t = f(W_{to} - W_{fuel}) \cdot sfc \cdot t \quad (1)$$

Fuel Weights Modelling

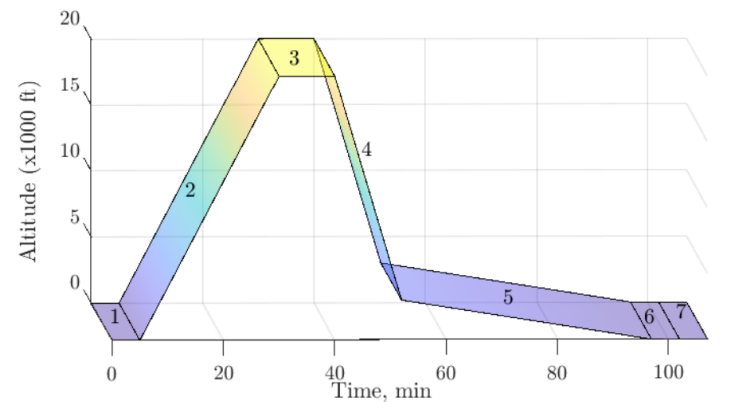


Figure 4: SynchroCapture's mission profile.

An overview of the mission profile in segments is shown in Figure 4, taken from the relevant Business Case report [10]. The mission profile was divided into discrete mission segments. Fuel weight losses from previous mission segments were taken into account when calculating mid-weight and maximum power at the start of each mission segment. The non-linear decrease in fuel weight during axial climb and descent was modelled using linear segmentation, by splitting them into multiple smaller segments and considering fuel weight losses at the start of each discretised segment.

Segment 1-2 (axial climb) was modelled to be vertical climb to 3000ft, forward flight for 75km, and then vertical climb to 20000ft, as defined during the preliminary weight sizing stage of the project. Vertical climb from 3000ft to 20000ft was discretised into 7 smaller individual segments, each with fuel weight subtracted from the previous, to account for power losses hence slower climb rate with increasing altitude. Segment 3-5 (axial descent), was also discretised into 7 smaller segments to account for decreasing altitude hence increasing power. Power required for a slope descent was modelled to be the average of power required for forward flight and vertical descent of the same segment [9]. Figure 5 presents the fuel weight consumed alongside vertical climb rate assumed during fuel weight estimation, in relation to time passed since the start of the mission.

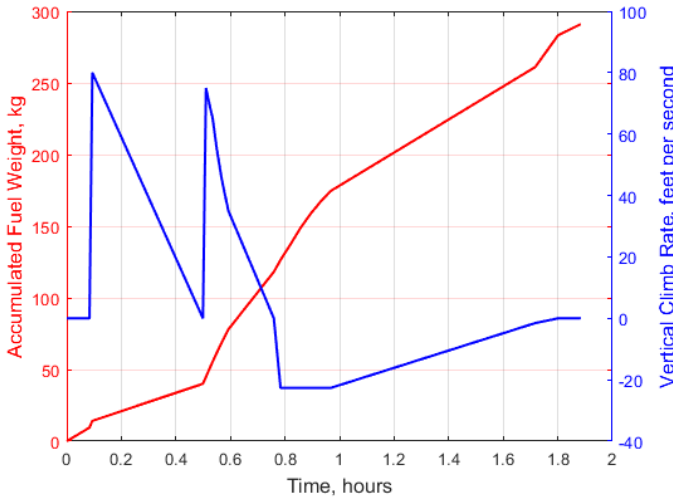


Figure 5: An overview of accumulated fuel weight required and vertical climb rate with respect to time passed in the mission.

The number of discretisation for a mission segment was mainly based on the change in climb rate, and was decided in collaboration with Arangath, O. [9]. Power calculations were utilised to obtain the maximum achievable climb rate of each discretised section, and a segment was split for each climb rate difference of about 10 feet per second. This resulted in 7 discretised sections mentioned earlier for both axial climb and descent segments with a consistent split of about 3000ft throughout.

Table 2: Summary of fuel weights allocation and aircraft weights.

Mission Segment	Fuel Weight (kg)	Aircraft Weight (kg)
0 - 1	9.5	1507.0
1 - 2	68.5	1497.5
2 - 3	40.2	1429.0
3 - 4	56.4	1388.8
4 - 5	86.5	2732.4
5 - 6	22.1	2645.9
6 - 7	7.8	2623.8
Total Fuel Weight	291.0	

A summary of fuel weights required and SynchroCapture's weight at the start of each mission segment is shown in Table 2. It should be noted that a relatively high sfc of 0.63 lb/h.p. per hour was chosen to overestimate the vehicle's fuel weight, although the final sfc was determined to be 0.55 lb/h.p. per hour. The increase in aircraft weight at segment 4-5 is also due to payload being captured in segment 3-4.

In the later stages of the project, power requirements were improved with the completion of the BEMT code and change in mission trajectories [10]. The climb rates and fuel weights for each mission segment were also re-calculated after a detailed analysis of SynchroCapture's flight performance, therefore both power and fuel weights were updated to more accurately represent the vehicle's performance and weight. They can be found in the related Aeroelasticity Modelling and Flight Performance Analysis reports [11, 12] respectively. A summary of SynchroCapture's weight breakdown immediately after the first convergence of MTOW can be found in [3], and the final updated weight breakdown can be found in Appendix A.1.

2 Fuel System

2.1 Regulations

Fuel tanks must be designed to retain its contents under the following loads - 1.5g Upward, 4g Forward, 2g Sideward, 4g Downward, according to FAR 29.952.

2.2 Fuel Tank Specifics & Placement

Fuel tanks are installed in a section of the helicopter structure designed to accommodate fuel. Consumable fuel of about 291 kg accounts for approximately 10% of the MTOW. The system includes two main tanks which hold 77.8 U.S. gallons (294.5 Litres) in total, 38.9 gallons each. The main tanks incorporate a series of baffles designed to mitigate fuel sloshing [13] without trapping fuel, and to stop the fuel surging during manoeuvres which would make the aircraft unstable and difficult to control.

An additional 10% was taken over fuel volume to determine the total volume of the fuel tanks required [14], to air cushion the fuel in the event of a hard landing when fully loaded, in compliance with FAR 29.952. This factors in the fuel tank wall thickness and the fuel absorption nature of the fuel tank sealants and valve sealants. There is also a substantial variation of fuel density with temperature. Density is lower on a hot day, so fewer kilograms of fuel will fit into the same tank, reducing the aircraft's range despite having full tanks. Cold fuel pumped into the tanks will also warm up and expand. This 10% safety factor provides ullage for a fuel expansion space, giving a total of $0.324 m^3$, $0.162 m^3$ for each fuel tank.

The main fuel tanks are designed with a low sump to accommodate various flight orientations, and are located at the longitudinal CG location of the helicopter but separated at an equal distance with respect to the lateral and vertical axis, as shown in Appendix A.2. This is because the rotor is located directly at the CG vertical axis and so is the slung load after capture, hence the presence of a load path and the requirement of a beam in between the main fuel tanks to support this load case. This distribution minimizes CG travel during the duration of flight and reduce fuel sloshing in the tanks. The splitting of main fuel system into front and back independent halves also ensures the fuel system is doubly redundant. The exact CG positions of

the tanks can be found in the related Fuselage Structural Design report [15].

Self-sealing bladder tanks are selected to stop leaks in the event of a crash. They are reinforced rubberised bags, rolled up and installed into the compartment through the fuel filler neck or removable access panel, and is secured by means of metal buttons or snap fasteners [16] inside the compartment.

2.3 Fuel System Configuration & Components

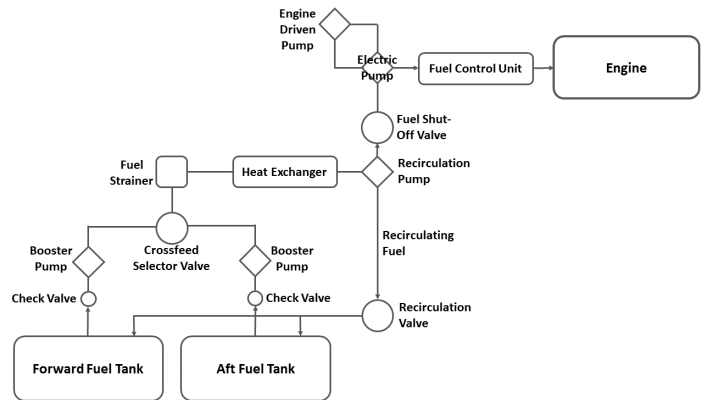


Figure 6: Schematic of SynchroCapture's fuel system.

A schematics of SynchroCapture's fuel system is completed and shown in Figure 6. Electric booster pumps are provided inside each main fuel tank. The fuel tanks are cross connected to ensure even fuel levels. Booster pumps supply fuel from the main fuel tanks to the engine fuel pump. Each main fuel supply line is equipped with a check valve ensuring fuel does not gravity feed from the booster pump back into the tanks and cause the booster pumps and lines to lose their prime [17]. Electrically Operated Crossfeed Selector Valve joins the system in a crossfeed way to ensure maximum fuel utilization. The fuel is piped from the tanks through fuel lines to this valve, which allows fuel to be fed to the engine from either the forward tank or the aft tank individually or from both at the same time. This facility allows the the fuel tanks to be balanced, trimming the aircraft longitudinally to maintain a desirable CG position, although no fuel exchange occurs between fuel tanks during flight. The fuel shut-off valve is capable of stopping fuel flow to the engine completely in the event of an emergency or fire [18] or to prevent unwanted fuel migration when the aircraft is not in

operation. This valve remains open at all normal operations and is located downstream before the electric pump.

A fuel filter strainer is incorporated to prevent foreign objects from entering the fuel system. It also removes moisture and other sediment from the fuel before it reaches the engine [18]. These contaminants are usually heavier than fuel and settle to the bottom of the sump where they can be drained out. Between the pump discharge and combustor fuel nozzles, fuel is routed through a heat exchanger to cool the engine oil and various subsystems [19]. An electrical fuel pump is used to maintain positive fuel pressure to the engine driven fuel pump and also serve as a backup in the event of mechanical pump failure. The engine driven pump is the primary pump that supplies fuel to the engine and operates any time the engine is running [18].

The essential control is provided immediately after the fuel has passed through the engine driven fuel pump [17], before being directed to the engine. The fuel control unit is part of the monitoring system detailed in section 3 (FADEC), and is between the engine driven fuel pump and combustion chambers. The flow range of the main fuel pump is related to the speed range of the pump drive system, contained in the engine accessory gearbox. Because of the relatively small variation gearbox drive speed, compared to the huge difference in fuel flow at different power requirements, delivered fuel flow can never match required fuel flow at all power settings. Thus, the engine fuel system encompasses a re-circulation loop that takes flow in excess back to the tanks [19].

2.4 Carburetor vs EFI

The carburetor, or alternatively an Electronic Fuel Injection (EFI) system, prepares the air-fuel mix in the inlet pipe within a narrow mixture margin. This certain mass ratio, called stoichiometric, is about 1:15 fuel burn in air [20], and is the theoretical ratio that leads complete combustion without leaving behind any hydrocarbon or oxygen. The EFI was ultimately selected for the system as it eliminates the problem of carburetor icing and the need for a carburetor heat system [21].

3 Health and Usage Monitoring System

As SynchroCapture is a completely new type of aircraft, safety and reliability are crucial points for its success. A crash in the early stages of introduction to the market would cost huge brand damage in a business point of view. Health and Usage Monitoring Systems (HUMS) are used to provide diagnostic information through a network of sensors to continuously monitor and analyse aircraft conditions to determine the status of flight critical subsystems and the overall health of the aircraft. Because of the large number of flight critical dynamic components on SynchroCapture, it is important to diagnose any problems effectively in real time during flight. HUMS are used on many aircraft as benefits include reduced mission aborts, increased time between failures and an increase in mission reliability [22]. They can raise warnings about potential failures prior to them occurring, thus preventing potentially fatal events. There are three distinct maintenance operations required on helicopters:

1. **Hard time:** Preventive maintenance required at fixed intervals.
2. **On condition:** A maintenance less rigorous than hard-time inspections. Performed to inspect or replace selected suspect components before approving the aircraft for continued operations.
3. **Condition monitoring:** A non-preventative process where the condition and data of all systems and components are assimilated and monitored on a continuous basis in order to apply corrective measures whenever necessary.

A layout of the HUMS aboard SynchroCapture can be found in Appendix A.3, and consists of on-board monitoring and off-board diagnostics and maintenance.

On-board

The on-board monitoring system gathers and relays information directly to the remote pilot and autopilot. This data include essential in-flight specification, such as engine oil temperature, average rotor torque usage and flight regime characteristics, and is critical to the proper operation of the vehicle. They are typically not heavily processed and may be easily condensed, hence

does not usually require large bandwidth or computational power. This data can be streamed to the ground station for use by an operator or integrated on-board for autonomous control.

Off-board

The off-board operator records relevant data from various sensors on board the aircraft during flight operations to the ground station's flight data recorder. This information is required for determining future necessary hard-time overhaul of components, and is interrogated automatically with a software that tracks the physical and temporal limits of the mechanical components. It also allows for a comprehensive life-cycle analysis to be conducted in real-time, thereby minimising maintenance downtime [23] and improving the safety of the vehicle dramatically. HUMS can notify ground crew of a component approaching its life expectancy and creates a diagnostic report that can be used by ground engineers.

3.1 Main rotor & hub

Continuous monitoring of the rotor system health is vital to the operation of any autonomous aircraft, and it is desirable to maintain optimal performance and maximize SynchroCapture's lifespan. A series of strain gauges and accelerometers are added on the rotor system to continually provide vibration and loads data to HUMS. The in-flight data from these sensors are compared with a preexisting database of potential risks and failure modes collected during initial flight testing and component certification phase.

Strain gauges also provide data for detection of potential fatigue problems. If the observed sensor data exhibit characteristics that indicate a potential or imminent failure mode, or that a component has reached its life-cycle limits, the system immediately alerts the ground control unit and transmits updated instructions to the autopilot. Additionally, the system creates a diagnostic report for the ground crew to be examined by maintenance technicians. Routine condition checks, such as track and balance, are performed using an infrared (IR) camera in conjunction with the collected vibration data from strain gauges and accelerometers to ensure optimal rotor performance.

3.2 Engine

The demands on the engine control system are exceptionally severe due to the requirement of a constant rotor speed even under varying loading conditions for the overall aircraft operation. Therefore, the main engine is equipped with a full authority digital engine control (FADEC) system. The system records and transmits important information to HUMS, including specifications such as time spent above operational torque limits, average flight performance data, oil temperature, fuel flow rate control, and fault monitoring. The HUMS also monitors the health of the FADEC that govern the engine fuel flow and power, and can automatically override the EFI and ignition system in case of fault.

3.3 Generator

Internal thermocouples and hall sensors are installed in the generator to monitor temperature. Additionally, current and voltage input to the controller is monitored to identify any electric faults. The diagnostic and fault data from the controller is directly streamed to the ground control unit.

3.4 Gearbox

Accelerometers and tachometers are placed on the main rotor gearbox, mounted to each rotor shaft and are connected to a gear, to monitor input RPM, output RPM, and vibration levels. The vibrations on both shafts are continuously monitored and examined against the tachometer pulse trains. The recorded data can be processed to obtain the vibrations on the main rotor drive shaft. Any anomaly will be recorded, and preventative action will be taken by the flight management system if required. The oil temperature, pressure, and chip detectors are monitored by in-line inductive oil debris monitors for oil leaks and debris accumulations.

3.5 Airframe

Accelerometers and strain gauges are embedded into the critical locations of the airframe and at critical stress points of main load-bearing structures throughout the aircraft, such as the gearbox housing, to monitor lateral and vertical accelerations ensuring accept-

able levels of vibration and determine faults within the avionic attitude sensors. This can diagnose any initiation and propagation of cracks in such members. Optics fibres along with Bragg gratings are used to determine strain, temperature, and deflections of critical airframe components. Since mechanical damage such as fractures, cracks, or delamination to the airframe destroys the fibres and interrupt the light flow, this anomaly can be quickly isolated.

3.6 Slung Line

Another important component to be monitored is the slung line, as it is key to the success of the mission. This can be achieved by introducing an intra-core fibre Bragg grating system. A variation in the environmental parameters including temperature and strain, influence both the pitch and the refractive index of the grating layers perturbing the spectral properties of the FBG. This perturbation in the spectrum is then utilized for sensing purposes [24].

As ropes are produced using rotating machinery, monitoring fibres can be helically wrapped around the internal structures of the rope as shown in Appendix A.4, modified from [25]. Distributed fibre optic sensor system using Bragg gratings that are written in the fibre during the drawing process prior to coating are used to identifying sections of the rope which show larger strains than the mean strain [25]. These areas would most likely suffer from rapid deterioration and will be subjected to visual inspections after each flight, as described in section 3.6.2.

3.6.1 Monitoring Fibre Sizing

The design of this fibre optic monitoring system focuses more on the rope than the grating sensor design, and was performed in the guidance of Monitoring Techniques [25]. The aspects considered in this design consist of strains experienced by the optical fibres and the curvature of the helix which the fibre optic is wound in. As the axial strain of the slung line is extremely small [2], it was therefore not a constraining design factor, since optical fibres have axial tensile failure strains of over 1% [26].

The fibre was sized using geometrical relations. Large number of turns around the rope with low attenuation are the desired characteristics of the fibres, as an optimal monitoring system covers large areas of the rope

whilst minimizing loss in signal light intensity. Therefore, a fine balance between the number of turns and total induced attenuation was chosen. The length ratio between the monitoring fibre and the rope, the strain ratio of the fibre within the rope to the overall strain of the rope, and the radius of curvature of the fibre [25], were defined by Equation 2, Equation 3, and Equation 4 respectively.

$$\frac{L_f}{L_r} = \frac{1}{LL_f} \sqrt{(\pi d_f)^2 + LL_f^2} \quad (2)$$

$$\frac{\epsilon_f}{\epsilon_r} = \cos^2 \alpha \quad (3)$$

$$\rho = \frac{\frac{d_f}{2}}{\sin^2 \alpha} \quad (4)$$

The fibre helix diameter must not exceed the rope diameter of 0.011 metres, thus it was first arbitrarily chosen to be 8 millimeters. The number of turns can be obtained by evaluating $\frac{L_r}{LL_f}$. An induced attenuation of 0.05 dB per 100 turns was defined to obtain the total attenuation. This is the value specified in Corning [27] for a radius of curvature of 37.5mm, and was assumed to be true for the range of radii around 37.5mm throughout the calculations. These relations mentioned above were then visualised in Figure 7 in order to determine the lay length of the fibre.

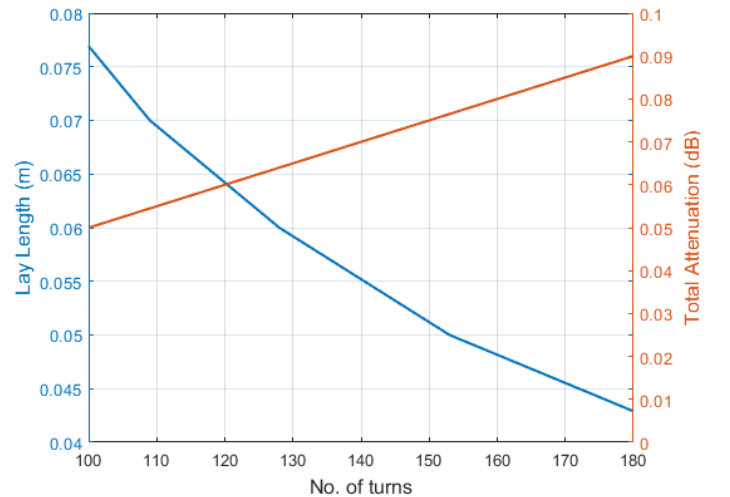


Figure 7: Lay length of fibre and total induced attenuation with respect to number of turns around the rope.

In the end, an arbitrary value of 120 number of turns was chosen, giving a lay length of 60 mm and attenu-

ation of 0.064 dB, which is within an acceptable level. A summary of fibre specifications is shown in Table 3. An overview of the design point in relation to the strain and length ratios can be found in Appendix A.5. With a fibre length of 8.26m, its weight was determined to be 0.023kg, assuming an off the shelf weight per km specified by Corning [28], which is an insignificant weight addition to the slung line.

Table 3: *Summary of fibre parameters.*

Parameter	Value
Optical fibre helix diameter (mm)	8
Lay length (mm)	60
Radius of curvature (mm)	27
Fibre rope strain ratio	0.851
Fibre rope length ratio	1.080
Induced attenuation (dB)	0.064
Axial strain of slung line (%)	1.69×10^{-3}
Axial strain of fibre (%)	1.44×10^{-3}

3.6.2 Rope Inspection & Retirement

Initial work was done in attempts to model the Dyneema rope lifetime using comparable logarithmic formulas, for example the modified Feyrer equation [29]. However, it was decided that the best way to monitor the health of the rope would be to conduct actual rope inspections after each use.

Factors like load history, bending radius, or a combination of factors, make retirement decisions difficult. Therefore, coupled with the monitoring FBG technology, rope inspection can be made easier than a strenuous process of observation before, during, and after each use. A visual inspection checklist includes abrasion due to filaments breaking, glazed areas caused by heat damage, discolouration due to chemical contamination, inconsistent diameter caused by overloading, and inconsistent texture due to shock load damage [30].

3.6.3 Limitations

Although FBG technology is attractive for distributed strain and temperature sensing and is fairly common in structural health monitoring (SHM), there are several limitations and challenges of implementing fibre optics for strain monitoring to overcome in its current stage.

1. There are currently no clear certification procedures to assess them.
2. Its applications to commercial aircraft is still in the research stage due to high stakes. The reliability of diagnosis under real operational and environmental conditions is somewhat questionable.
3. Applying off the shelf solutions directly to a completely new type of aircraft structure is a bit of a stretch.
4. The slung line was selected to be an off the shelf Dyneema rope [2], hence the option of a purpose produced FBG embedded rope might require significant extra cost.
5. It is only capable of measuring strain at discrete points along the fibre optic, where the gratings are positioned, thus would not provide sufficient resolution. Multiple fibres would be required to form various configurations of the distribution of Bragg grating sensors.

4 Combustion Systems

Work presented in this section was done after engine compressor and turbine sizing. A summary of parameters at their respective engine stations is presented in Appendix A.6, and is detailed in the related report [9]. These values were necessary to complete the sizing of the engine annular combustor. It should be noted that the design of combustion systems was accomplished in the guidance of Mattingly [19], Chapter 9.

4.1 Layout

The inner, outer radii and radial heights at stations 3.1 and 4 were calculated using the annular geometry relations provided. Two different design options, A and B, were then defined, by matching the radius of burner outer casing to the outer radii at station 3.1 and 4 correspondingly. Annular geometries and the number of fuel nozzle assemblies were then calculated for both designs. A summary of the parameters is shown in Table 4 and the two designs were visualised in Figure 8. Design B was ultimately chosen over A as it has fewer fuel nozzles, although mixing in the secondary and dilution

zones are more effective in design A due to smaller radial height and lengths of secondary and dilution zones (L_{sz} and L_{dz} are approx. proportional to H).

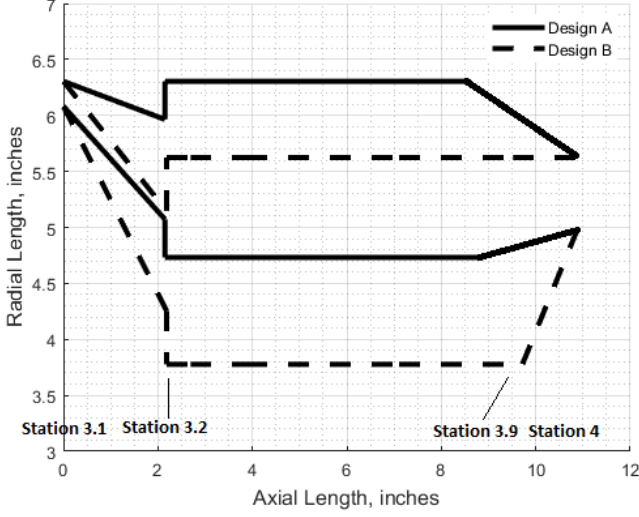


Figure 8: Layouts of two different design choices.

Table 4: Summary of two possible burner layouts.

Parameter (inches)	Design A $r_{o3.2} = r_{o3.1}$	Design B $r_{o3.2} = r_{o4}$
r_o	6.305	5.624
r_i	4.732	3.778
H	1.573	1.846
$N_{noz}(\text{dimensionless})$	23	16

4.2 Diffuser

A diffuser is required to slow down high speed flow from the compressor, in order to achieve a low Mach number at the burner inlet (station 3.2) to promote efficient combustion and avoid large total pressure losses [31]. Mach number at burner inlet was chosen to be 0.05 for a subsonic aircraft [32], thus the diffuser area ratio can be determined using 1-D compressible flow relations [33] and Equation 5, knowing the Mach numbers at the respective stations.

$$AR = \frac{A_{3.2}}{A_{3.1}} = \frac{\left(\frac{A}{A^*}\right)_{M=0.05}}{\left(\frac{A}{A^*}\right)_{M=0.32}} \quad (5)$$

Single splitter plate was originally chosen for the annular diffuser as a compromise between geometric com-

plexity and the length of the diffuser, as the number of parallel streams equal to the reduction factor of the diffuser length, giving a two-stream optimal flat-wall diffuser. However, the diffuser area ratio ($AR = 6.3$) is greater than the maximum target of 4, hence a dump diffuser is added onto the flat-wall diffuser to improve its performance. The dump diffuser acts as a flow straightening section, and was sized to allow sufficient axial distance for the separated flow at dump diffuser entry to reattach to the upper and lower walls [19]. A summary of the combined diffuser parameters is shown in Table 5. Figure 9 presents the reduction in diffuser length by employing a splitter plate, as it gives additional diffusion action.

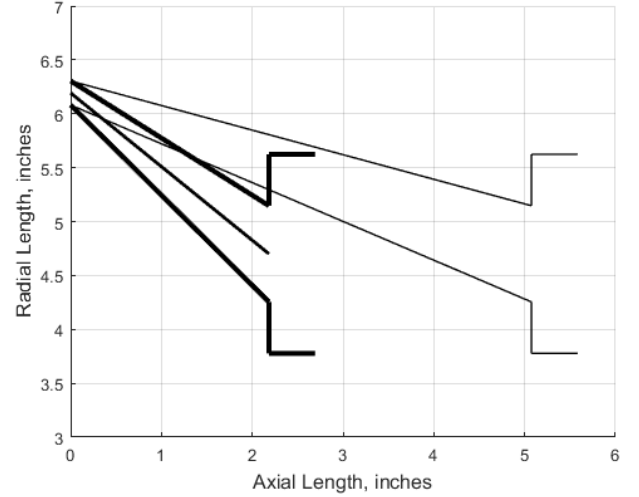


Figure 9: Layout for two diffuser designs, without and with a splitter plate.

Table 5: Summary of the combined diffuser parameters.

Parameter	Value
AR	6.282
$A_{3.2} \text{ (ft}^2\text{)}$	0.379
$L_d \text{ (inches)}$	0.509
$L_m \text{ (inches)}$	2.645
Flat-wall diffuser axial length (inches)	2.184
Combined diffuser axial length (inches)	2.693

4.3 Air Partitioning

The liner gas temperature, T_g , was determined by studying its relations with the air mass fractions for

primary and secondary zones, μ_{pz} and μ_{sz} , as well as the cooling mass fraction, μ_c . Figure 10 presents these relations by visualising the allocation of mass flow fractions with varying liner gas temperature.

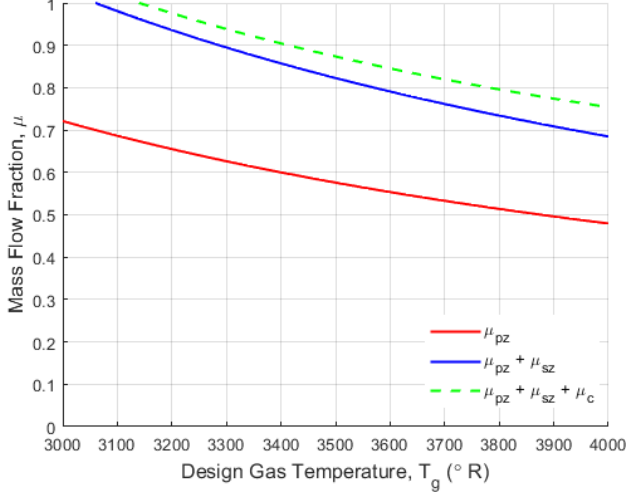


Figure 10: Variation of mass flow fractions with respect to design gas temperatures.

Considering the minimisation of air pollutant emissions, a good design target for T_g is about 3200 °R. This temperature was selected because it gave a primary zone equivalence ratio of 0.624, which is above the lean blowout limit of 0.5, and also sufficient mass fraction allocation for both primary and secondary zones as well as transpiration liner cooling in addition to extra mass fraction allocated for the dilution zone. A summary of the air partitioning is shown in Table 6.

Table 6: Air partitioning for the engine burner.

Burner Zone	Air Flow (kg/s)	Mass Fractions
Primary	2.23	0.624
Secondary	1.00	0.281
Transpiration Cooling	0.14	0.040
Dilution	0.09	0.024
Total	3.57	1.000

4.4 Dome and Liner

Using isentropic flow relations from Essentials [34], static and total pressure at station 3.2 can be obtained using the total pressure and temperature at station 3.1,

stated in A.6. The liner height was then found to be 1.613 inches using equations provided in [19]. The allocated total pressure loss was also examined to ensure it provides suitable stirring by the primary zone swirler. The minimum required total pressure loss coefficient was found to be 20.4% of the allocated main burner pressure loss coefficient, therefore it is within a comfortable margin of less than 40%.

4.5 Combustion Zones Sizing

All parameters and dimensions of the primary, secondary and dilution zone were sized using various data acquired, and presented in Table 6. The swirler was also sized to ensure it fits within the liner. An overview of the combustor layout is presented in Figure 11.

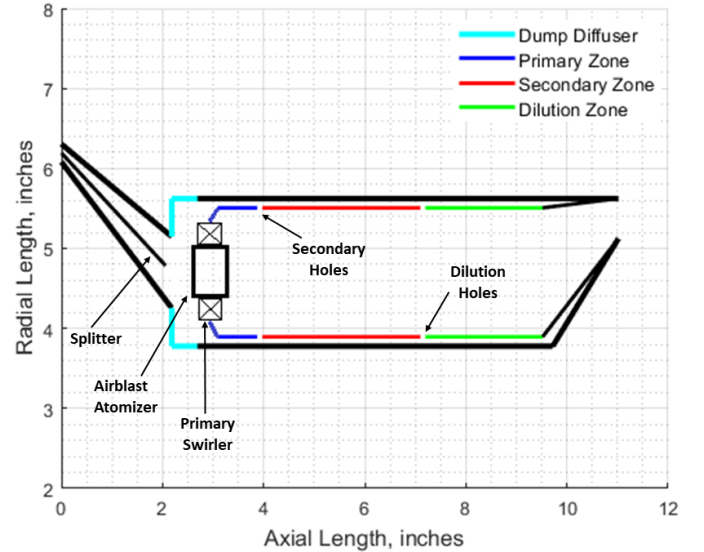


Figure 11: A 1:1 scale view of the engine burner layout.

Table 7: Summary of burner axial lengths and main parameters.

Parameter	Value
Primary Zone axial length (inches)	1.181
Number of Secondary Holes	912
Secondary Zone axial length (inches)	3.225
Number of Dilution Holes	15
Dilution Zone axial length (inches)	2.419
Total length of main burner liner (inches)	6.824
Total length of main burner (inches)	9.517

5 Conclusion

After 5 weeks of project a final design which satisfies most of the requirements was obtained. Lots of changes were made during the whole duration of this period, specifically the component and fuel weights over the past two weeks. However, due to the excellent communication between the different teams, individual designs and calculations were able to adapt to the requirements. Decisions made were always well informed to the entire team.

Preliminary MTOW of 2907 kg differs about 0.52% from the final MTOW of 2922 kg evaluated as shown in A.1, demonstrating exceptional accuracy of preliminary estimations. Notable changes in component weights include the under-estimations of fuselage and empennage group, without taking into account the tail mechanism and ballast added in the later stages of the project, and the over-estimation of avionics group, in particular UAV avionics. A fuel system was successfully designed with the capability of balancing fuel CG. A typical combustion chamber with an annular configuration is also designed. Although much work is needed to be done on the FBG slung line monitoring system, the overall project is completed with much satisfaction.

6 Future Work

Limitations and shortcomings of the presented work are discussed throughout, and future work can be done on all aspects mentioned in this report. This includes:

1. The weight sizing algorithm could be repeated for various combinations of vehicle parameters. Trade studies could be conducted to examine the merits of varying aspect ratio, rotor tip speed, and disc loading. This would take into account the MTOW as one of a deciding factor on the rotor and tail parameters, instead of being just an end product of these fixed inputs.
2. Final total fuel weight is actually 60 kg lower than initial total fuel weight estimation presented above. Although an over prediction was to be expected, a fuel dump system could be designed to get rid of the excess fuel in the events of an emergency before returning to the ship, or when weight reduction is required.
3. FBG sensors are not a well-established research field, but a large market awaits due to its sensitivity and low costs [24]. Although it is currently only an idea proposed for rope monitoring, a lot more research could be conducted just on fibre optics itself. The precision of the grating sensing sites to the physical position along the length of the rope could also be optimised to monitor majority of the rope area.
4. The possibility of innovating a new design of the combustion chamber instead of the annular design is unlikely, although the dump diffuser could be optimised to better counter the occurrence of flow separation effects. One example is to introduce vortex generators or some sort of protrusions to delay separation.

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A Appendices

A.1 SynchroCapture's Weight Breakdown

Table 8: Summary of final weight breakdown. Note MTOW includes payload and is not the actual weight during take-off.

Component	Value (Metric, kg)	Value (Imperial, lb)
Rotor Group	165.8	365.5
Blades	98.3	216.7
Hub	49.5	109.0
Spinner	18.0	39.7
Empennage Group	63.5	140.0
Horizontal & Vertical Tails	25.1	55.2
Tail Boom	15.5	34.1
Tail Mechanism	23.0	50.6
Fuselage Group	100.0	220.0
Lighting Gear Group	30.0	66.0
Propulsion Group	268.1	591.1
Engine	207.3	456.6
Accessories	34.5	76.1
Nacelle	26.4	58.2
Drive System Group	221.7	488.8
Gearbox	188.1	414.7
Rotor Shafts	28.1	62.0
Drive Shaft	0.7	1.6
Rotor Brake	4.7	10.4
Fuel Systems Group	55.0	121.3
Avionics Group	150.0	330.7
Ballast	35.0	77.2
Flight Controls Group	80.0	176.4
Fluids (oil, unusable fuel)	23.0	50.7
Anti-Icing Systems	39.0	86.0
Empty Weight	1231.1	2714.1
Fuel	291.0	641.5
Payload (Slung Load)	1400.0	3086.5
Maximum Take-Off Weight	2922.1	6442.1

A.2 Diagram of Fuel Tanks Placement

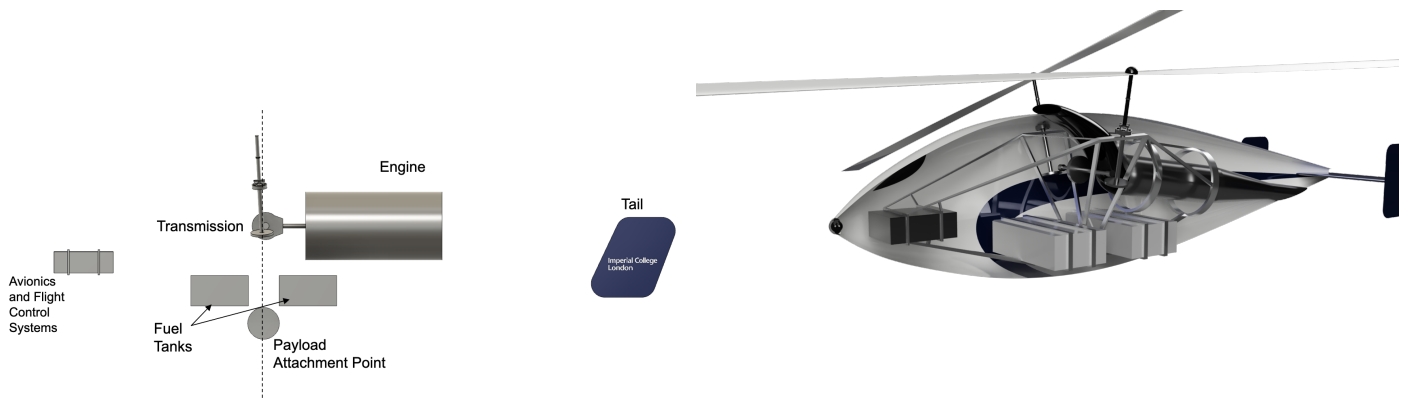


Figure 12: Fuel tanks placement in relation to other components within SynchroCapture. Note the dashed line in the image on the left is the vertical CG axis.

A.3 HUMS Layout

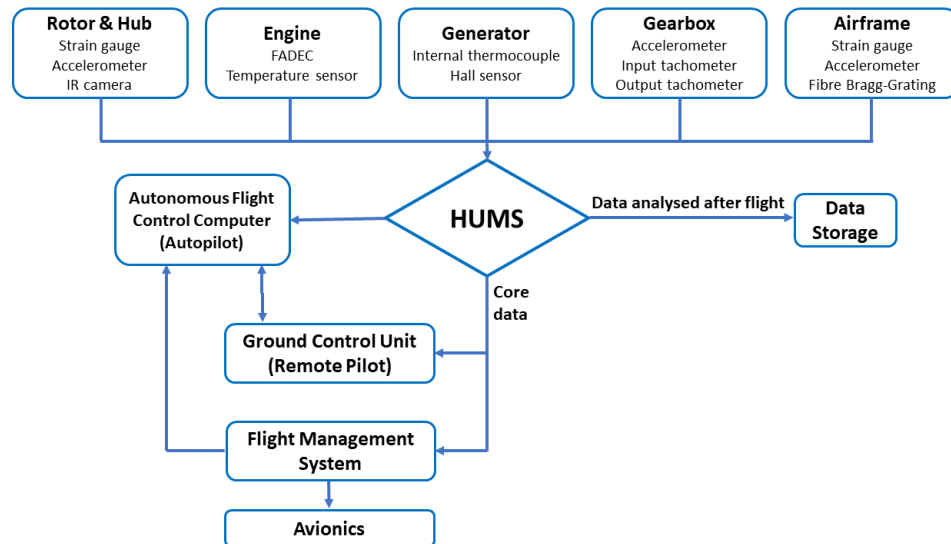


Figure 13: HUMS layout.

A.4 Monitoring Fibre Helix

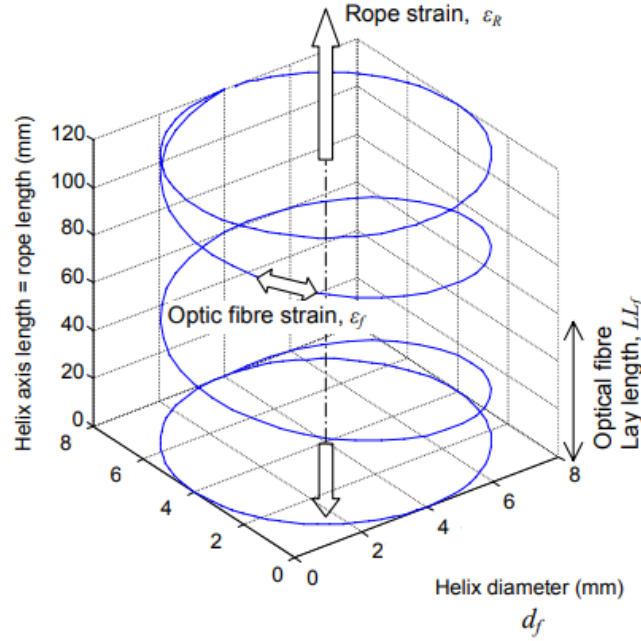
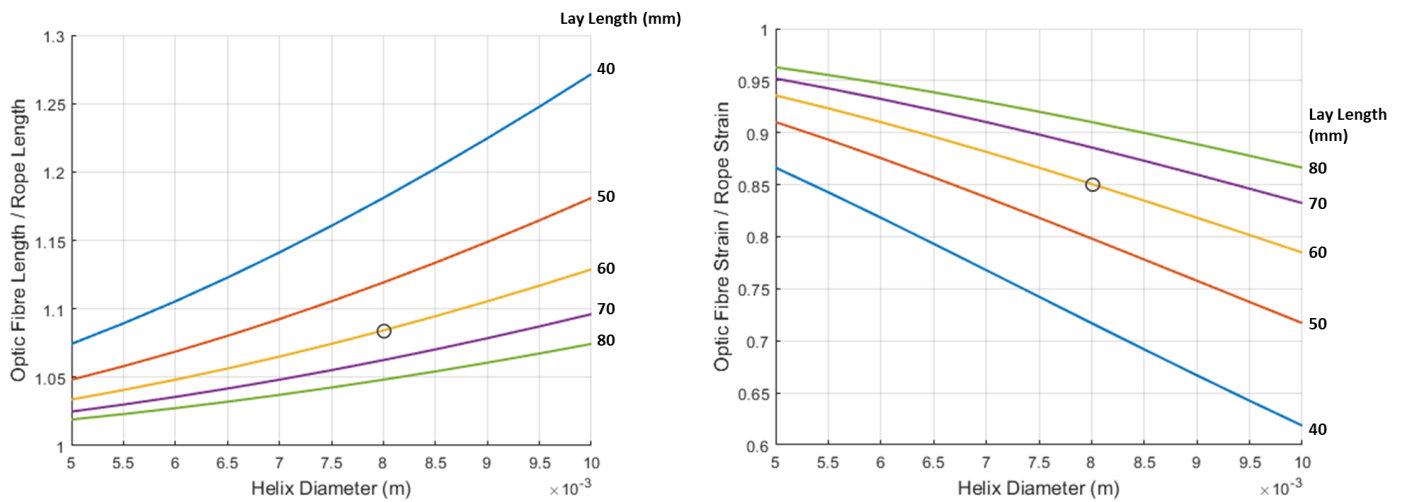


Figure 14: Helical path of the optical monitoring fibre within the rope.

A.5 Optical Fibre Design Points



A.6 Engine Stations

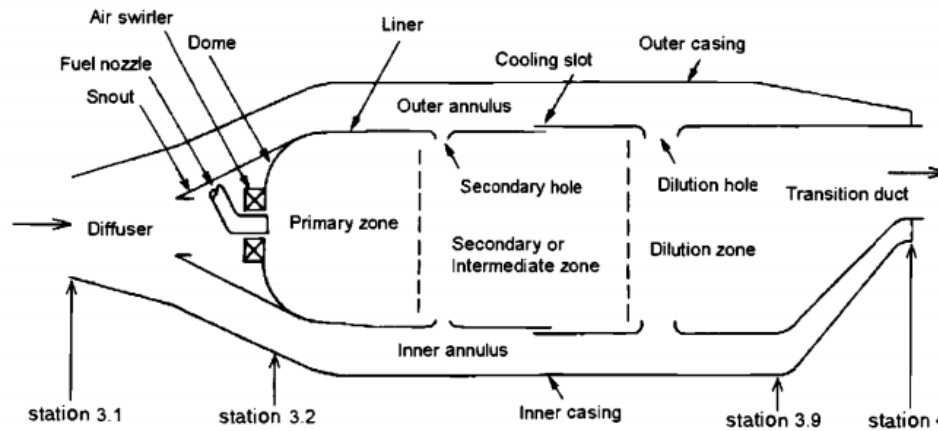


Figure 16: Principal features and main stations of the burner, obtained from Mattingly.

Table 9: Summary of compressor and turbine parameters.

Station	Mass flow rate (kg/s)	P_T (kPa)	T_T (K)	Mach	Mean radius (m)	Area (m^2)
3.1	3.57	982.9	565	0.32	0.1573	0.0056
4	3.38	904.0	1700	0.25	0.1393	0.0062

B Supporting MATLAB Code

B.1 FuelEstimate.m

This script evaluates fuel weight estimations and plots the relevant graphs.

```

1
2 clear
3 clc
4 close all
5
6 % Script written by Jo Wayne Tan, to give estimates of
7 % power required and fuel weights for Purpose Built Helicopter. 22/5/2020
8 % Based on Maryland Reports, Preliminary Vehicle Sizing
9
10 %% Fixed values
11 rho_0 = 1.225; % Air density at sea level
12 rho = 0.653; % Air density at 20,000 ft
13 g = 9.81; % Gravity
14 z_mrbl = 4; % Number of main rotor blades
15 W_pl = 1400; % Payload weight
16
17 %% Design Drivers
18 % Target Geometric Inputs
19 DL = 9.4; % Disc loading, DL = W_to/disc_area
20 AR = 14; % Blade reference aspect ratio

```

```

21 sigma_mr = 0.0909;          % Main rotor solidity, sigma_mr = W.to/((C.t.sigma_mr) * rho * pi ...
    * R_mr^2 * omega_R_mr^2) from Tishchenko
22
23 % Target Performance Parameters
24 FM_mr = 0.6667;             % Figure of merit, main rotor
25 eta_pr = 0.98;              % Main rotor propulsive efficiency
26
27 %% Power (mission segment requires most power used to size installed power)
28 % Inputs
29 W.to = 2907;                 % Initial MTOW guess – with payload          (CHANGE FOR EVERY ITERATION)
30 R_mr = 13.2;                 % Main rotor radius, ft (3.28)              (CHANGE FOR EVERY ITERATION)
31 disc_area = pi * R_mr^2 * 2; % Area of the rotor disc (2 rotors)
32
33 %% Fuel for segment 0-1
34 % Input of W.segment and time depends on mission segment
35 W.segment01 = (W.to - W.pl) * 2.20;
36 rho_01 = 23.77 * 10^(-4); % Air density at 0 ft
37 time01 = 5/60;              % Time taken, hours
38 DiscLoading = 5;
39
40 [P.mw01, W.fuel01] = fuelest(W.segment01, rho_01, disc_area, time01, 'Hover', 0, 0, 0, ...
    DiscLoading);
41
42 disp(W.fuel01)
43
44 %% Fuel for segment 1-2 (climb 0 to 3000 ft)
45 W.segment02a1 = (W.to - W.fuel01 - W.pl) * 2.2; % weight after consuming fuel for segment 0-1
46 rho_02a = 21.796 * 10^(-4); % Air density at 3,000 ft
47 C02a = 80;                   % feet per second
48 time02a1 = (3000/C02a)/(3600); % Time taken, hours
49 DiscLoading = 5;
50
51 [P.mw02a1, W.fuel02a1] = fuelest(W.segment02a1, rho_02a, disc_area, time02a1, 'VClimb', ...
    3000, 0, C02a, DiscLoading);
52
53 %% Fuel for segment 1-2 horizontal flight
54 % (probably wanna travel at ~3000ft and then climb?) (16 mins)
55 W.segment02a2 = W.segment02a1 - W.fuel02a1 * 2.2;
56 rho_02a = 21.796 * 10^(-4); % Air density at 3,000 ft
57 time02a2 = 24.324/60;        % Time taken, hours 32.43
58 V02 = 185 * 0.539957;        % in kts
59 DiscLoading = 5;
60
61 [P.mw02a2, W.fuel02a2] = fuelest(W.segment02a2, rho_02a, disc_area, time02a2, 'Flight', ...
    3000, V02, 0, DiscLoading);
62
63 %% Fuel for segment 1-2 vertical climb (3000 ft division)
64 W.segment02b1 = W.segment02a2 - W.fuel02a2 * 2.2;
65 rho_02b1 = 19.896 * 10^(-4); % Air density at 6,000 ft
66 C02b1 = 75;                   % feet per second
67 time02b1 = (3000/C02b1)/(3600);
68 DiscLoading = 5;
69
70 [P.mw02b1, W.fuel02b1] = fuelest(W.segment02b1, rho_02b1, disc_area, time02b1, 'VClimb', ...
    6000, 0, C02b1, DiscLoading);
71
72 W.segment02b2 = W.segment02b1 - W.fuel02b1 * 2.2;
73 rho_02b2 = 18.144 * 10^(-4); % Air density at 9,000 ft
74 C02b2 = 70;                   % feet per second

```

```

75 time02b2 = (3000/C02b2)/(3600);
76
77 [P.mw02b2, W.fuel02b2] = fuelest(W.segment02b2, rho.02b2, disc.area, time02b2, 'VClimb', ...
    9000, 0, C02b2, DiscLoading);
78
79 W.segment02b3 = W.segment02b2 - W.fuel02b2 * 2.2;
80 rho.02b3 = 16.52 * 10^(-4); % Air density at 12,000 ft
81 C02b3 = 65; % feet per second
82 time02b3 = (3000/C02b3)/(3600);
83
84 [P.mw02b3, W.fuel02b3] = fuelest(W.segment02b3, rho.02b3, disc.area, time02b3, 'VClimb', ...
    12000, 0, C02b3, DiscLoading);
85
86 W.segment02b4 = W.segment02b3 - W.fuel02b3 * 2.2;
87 rho.02b4 = 14.96 * 10^(-4); % Air density at 15,000 ft
88 C02b4 = 55; % feet per second
89 time02b4 = (3000/C02b4)/(3600);
90
91 [P.mw02b4, W.fuel02b4] = fuelest(W.segment02b4, rho.02b4, disc.area, time02b4, 'VClimb', ...
    15000, 0, C02b4, DiscLoading);
92
93 W.segment02b5 = W.segment02b4 - W.fuel02b4 * 2.2;
94 rho.02b5 = 13.586 * 10^(-4); % Air density at 18,000 ft
95 C02b5 = 45; % feet per second
96 time02b5 = (3000/C02b5)/(3600);
97
98 [P.mw02b5, W.fuel02b5] = fuelest(W.segment02b5, rho.02b5, disc.area, time02b5, 'VClimb', ...
    18000, 0, C02b5, DiscLoading);
99
100 W.segment02b6 = W.segment02b5 - W.fuel02b5 * 2.2;
101 rho.02b6 = 12.67 * 10^(-4); % Air density at 20,000 ft
102 C02b6 = 35; % feet per second
103 time02b6 = (3000/C02b6)/(3600);
104
105 [P.mw02b6, W.fuel02b6] = fuelest(W.segment02b6, rho.02b6, disc.area, time02b6, 'VClimb', ...
    20000, 0, C02b6, DiscLoading);
106
107 W.fuel02 = W.fuel02a1 + W.fuel02a2 + W.fuel02b1 + W.fuel02b2 + W.fuel02b3 + W.fuel02b4 + ...
    W.fuel02b5 + W.fuel02b6;
108 time02 = time02a1 + time02a2 + time02b1 + time02b2 + time02b3 + time02b4 + time02b5 + ...
    time02b6;
109
110 disp(W.fuel02)
111
112 %% Fuel for segment 2-3
113 W.segment03 = (W.to - W.fuel01 - W.fuel02 - W.pl) * 2.2; % weight after consuming fuel ...
    for segment 0-2
114 rho.03 = 12.67 * 10^(-4); % Air density at 20,000 ft
115 time03 = 10/60; % Time taken, hours
116 DiscLoading = 5;
117
118 [P.mw03, W.fuel03] = fuelest(W.segment03, rho.03, disc.area, time03, 'Hover', 20000, 0, 0, ...
    DiscLoading);
119
120 disp(W.fuel03)
121
122 %% Fuel for segment 3-4 axial descent (3000 ft division)
123 W.segment041 = (W.to - W.fuel01 - W.fuel02 - W.fuel03) * 2.2;
124 rho.041 = 12.67 * 10^(-4); % Air density at 20,000 ft

```

```

125 C041 = -22.667;           % feet per second
126 V04 = 144 * 0.539957;    % in kts
127 time041 = ((12.5/60)/17)*2;
128 DiscLoading = 5;
129
130 [P.mw041, W_fuel041] = fuelest(W_segment041, rho_041, disc_area, time041, 'ClimbCruise', ...
    20000, V04, C041, DiscLoading);
131
132 W_segment042 = W_segment041 - W_fuel041 * 2.2;
133 rho_042 = 13.586 * 10^(-4); % Air density at 18,000 ft
134 C042 = -22.667;           % feet per second
135 time042 = ((12.5/60)/17)*3;
136
137 [P.mw042, W_fuel042] = fuelest(W_segment042, rho_042, disc_area, time042, 'ClimbCruise', ...
    18000, V04, C042, DiscLoading);
138
139 W_segment043 = W_segment042 - W_fuel042 * 2.2;
140 rho_043 = 14.96 * 10^(-4); % Air density at 15,000 ft
141 C043 = -22.667;           % feet per second
142 time043 = ((12.5/60)/17)*3;
143 DiscLoading = 9.5;
144
145 [P.mw043, W_fuel043] = fuelest(W_segment043, rho_043, disc_area, time043, 'ClimbCruise', ...
    15000, V04, C043, DiscLoading);
146
147 W_segment044 = W_segment043 - W_fuel043 * 2.2;
148 rho_044 = 16.52 * 10^(-4); % Air density at 12,000 ft
149 C044 = -22.667;           % feet per second
150 time044 = ((12.5/60)/17)*3;
151
152 [P.mw044, W_fuel044] = fuelest(W_segment044, rho_044, disc_area, time044, 'ClimbCruise', ...
    12000, V04, C044, DiscLoading);
153
154 W_segment045 = W_segment044 - W_fuel044 * 2.2;
155 rho_045 = 18.144 * 10^(-4); % Air density at 9,000 ft
156 C045 = -22.667;           % feet per second
157 time045 = ((12.5/60)/17)*3;
158
159 [P.mw045, W_fuel045] = fuelest(W_segment045, rho_045, disc_area, time045, 'ClimbCruise', ...
    9000, V04, C045, DiscLoading);
160
161 W_segment046 = W_segment045 - W_fuel045 * 2.2;
162 rho_046 = 19.896 * 10^(-4); % Air density at 6,000 ft
163 C046 = -22.667;           % feet per second
164 time046 = ((12.5/60)/17)*3;
165
166 [P.mw046, W_fuel046] = fuelest(W_segment046, rho_046, disc_area, time046, 'ClimbCruise', ...
    6000, V04, C046, DiscLoading);
167
168 W_fuel04 = W_fuel041 + W_fuel042 + W_fuel043 + W_fuel044 + W_fuel045 + W_fuel046;
169 time04 = time041 + time042 + time043 + time044 + time045 + time046;
170
171 disp(W_fuel04)
172
173 %% Fuel for segment 4-5
174 W_segment05 = (W_to - W_fuel01 - W_fuel02 - W_fuel03 - W_fuel04) * 2.2; % weight after ...
    consuming fuel for segment 0-4, with payload
175 rho_05 = 21.796 * 10^(-4); % Air density at 3,000 ft
176 time05 = 24.324/60;       % Time taken, hours

```

```

177 C05 = -1.5417;           % feet per second (-3)
178 V05 = 185 * 0.539957;    % in kts
179 DiscLoading = 9.5;
180
181 [P.mw05, W.fuel05] = fuelest(W.segment05, rho_05, disc_area, time05, 'ClimbCruise', 3000, ...
    V05, C05, DiscLoading);
182
183 disp(W.fuel05)
184
185 %% Fuel for segment 5-6
186 % Weight after consuming fuel for segment 0-5, with payload
187 W.segment06 = (W.to - W.fuel01 - W.fuel02 - W.fuel03 - W.fuel04 - W.fuel05) * 2.2;
188 rho_06 = 23.77* 10^(-4); % Air density at 0 ft
189 time06 = 5/60;           % Time taken, hours
190
191 [P.mw06, W.fuel06] = fuelest(W.segment06, rho_06, disc_area, time06, 'Hover', 0, 0, 0, ...
    DiscLoading);
192
193 disp(W.fuel06)
194
195 %% Fuel for segment 6-7
196 % Weight after consuming fuel for segment 0-6, without payload
197 W.segment07 = (W.to - W.fuel01 - W.fuel02 - W.fuel03 - W.fuel04 - W.fuel05 - W.fuel06 - ...
    W.pl) * 2.2;
198 rho_07 = 23.77* 10^(-4); % Air density at 0 ft
199 time07 = 5/60;           % Time taken, hours
200 DiscLoading = 5;
201
202 [P.mw07, W.fuel07] = fuelest(W.segment07, rho_07, disc_area, time07, 'Hover', 0, 0, 0, ...
    DiscLoading);
203
204 disp(W.fuel07)
205
206 %% Total Fuel Weight
207 W.tfuel = W.fuel01 + W.fuel02 + W.fuel03 + W.fuel04 + W.fuel05 + W.fuel06 + W.fuel07;
208 time = time01 + time02 + time03 + time04 + time05 + time06 + time07;
209
210 %% Plots
211 timedata = [0 time01 time02a1 time02a2 time02b1 time02b2 time02b3 time02b4 time02b5 ...
    time02b6 time03 time04a1 time04a2 time04a3 time04a4 time04a5 time04a6 0.75 time06 time07];
212 fueldata = [0 W.fuel01 W.fuel02a1 W.fuel02a2 W.fuel02b1 W.fuel02b2 W.fuel02b3 W.fuel02b4 ...
    W.fuel02b5 W.fuel02b6 W.fuel03 W.fuel04a1 W.fuel04a2 W.fuel04a3 W.fuel04a4 W.fuel04a5 ...
    W.fuel04a6 W.fuel05 W.fuel06 W.fuel07];
213 climbdata = [0 0 80 0 75 70 65 55 45 35 0 -22.667 -22.667 -22.667 -22.667 -22.667 -22.667 ...
    -1.5417 0 0];
214 timeplot = zeros(length(timedata));
215 fuelplot = zeros(length(fueldata));
216 climbplot = zeros(length(climbdata));
217
218 for i = 1:length(timedata)
219     timeplot(i) = sum(timedata(1:i));
220     fuelplot(i) = sum(fueldata(1:i));
221     climbplot(i) = climbdata(i);
222 end
223
224 figure (1)
225 hold on
226 grid on
227

```

```

228 [AX, H1, H2] = plotyy(timeplot, fuelplot, timeplot, climbplot, 'plot');
229 set(AX(2), 'YLim', [-40 100], 'YTickMode', 'auto')
230 xlim(AX(1), [0 2])
231 xlim(AX(2), [0 2])
232 set(H1, 'LineWidth', 1.5);
233 set(H2, 'LineWidth', 1.5);
234 set(AX, {'ycolor'}, {'r'; 'b'})
235 set(H1, 'Color', 'r');
236 set(H2, 'Color', 'b');
237 xlabel('Time, hours', 'FontSize', 12);
238 ylabel(AX(1), 'Accumulated Fuel Weight, kg', 'FontSize', 12);
239 ylabel(AX(2), 'Vertical Climb Rate, feet per second', 'FontSize', 12);

```

B.2 fuelest.m

This is the fuel function for the FuelEstimate.m script

```

1
2 function [P_mw, W_fuel] = fuelest(W_segment, rho, DiscArea, time, flightcase, altitude, V, ...
   Climbrate, DiscLoading)
3 % W_fuel = fuelest(W_segment, rho, DiscArea, time, flightcase, altitude, V, Climbrate, ...
   DiscLoading)
4
5 sfc = 0.55; % sfc in lb/h.p. hr (turbines), 0.63
6 % DiscLoading = 9.5; % lbs/ft^2
7 FuselageArea = 4.88 * 10.7639; % Fuselage area, ft^2 (ESTIMATE)
8 FigureOfMerit = 0.6667; % (ESTIMATE)
9 fuel_0 = 300; % Initial guess of fuel weight for segment, kg
10 err = 1; % tolerance for fuel weight to converge
11
12 while err > 0.01
13     switch flightcase
14         case 'Hover'
15
16             Thrust = GetThrustHov(W_segment - 0.5 * fuel_0, DiscArea, FuselageArea);
17
18             hpinduced = Thrust.*sqrt(DiscLoading./(2*rho))/550;%Get induced power
19
20             hptotal = hpinduced/FigureOfMerit;%Get total power
21
22             [P_mw, ~] = PowerLosses(hptotal, 15000, altitude);%Account for losses
23
24         case 'Vclimb'
25
26             Thrusthov = GetThrustHov(W_segment - 0.5 * fuel_0, DiscArea, FuselageArea);%Get ...
               required thrust to hover.
27
28             hpinduced = Thrusthov.*sqrt(DiscLoading./(2*rho))/550;%Induced power for hover
29
30             Thrust = GetThrustClimb(W_segment - 0.5 * ...
               fuel_0, DiscArea, FuselageArea, Climbrate, rho);%Get required thrust for climb
31
32             Deltahp = ClimbPower(Climbrate, Thrust, rho, DiscArea, W_segment - 0.5 * ...
               fuel_0, FuselageArea);%Calculate increase in required power
33
34             hptot = hpinduced + Deltahp;%Calculate total required induced power
35

```

```

36         hptotal = hptot/FigureOfMerit;%Calculate total power
37
38         [P_mw,~] = PowerLosses(hptotal,15000,altitude);%Account for losses
39
40         case 'Flight'
41
42             Thrusthov = GetThrustHov(W_segment - 0.5 * fuel_0,DiscArea,FuselageArea);%Get ...
                     thrust required for hover
43
44             hpinduced = FlightPower(V,Thrusthov,rho,DiscArea);%Induced power for hover
45
46             hptotal = hpinduced; %Total power required
47
48             [P_mw,~] = PowerLosses(hptotal,15000,altitude);%Account for losses
49
50         case 'ClimbCruise'
51
52             [P_mw] = ClimbCruise(W_segment - 0.5 * fuel_0, DiscArea, FuselageArea, ...
                     DiscLoading, Climbrate, rho, altitude, V);
53
54         end
55
56         W_fuel = P_mw * sfc * time;                % Fuel weight, lb
57         W_fuel = W_fuel * 0.453592;                % Fuel weight, kg
58         err = abs((W_fuel - fuel_0)/W_fuel)) * 100; % Percentage error between previous ...
                     and current iteration
59         fuel_0 = W_fuel;
60     end
61
62 end

```

B.3 Burner.m

This script evaluates burner parameters and plots relevant graphs.

```

1
2 clear
3 clc
4 close all
5
6 % Script written by Jo Wayne Tan to size combustion systems
7 % of Purpose Built Helicopter. 03/06/2020
8 % Main Burner Design based on Mattingly Methods
9
10 %% Unit conversions
11 m2ft = 3.28;
12 m2in = 39.3701;
13 ft2in2 = 144;
14 kg2lbs = 2.205;
15 kW2hp = 1.341;
16 m2ft2 = 10.764;
17 kpa2psia = 0.145;
18 r2k = 0.555556;
19
20 %% Inputs
21 A_31 = 0.0056 * m2ft2; % Area of station 3.1, ft^2
22 A_4 = 0.0062 * m2ft2; % Area of station 4, ft^2

```

```

23 r_m31 = 0.1573 * m2in; % Mean radii at station 3.1, in
24 r_m4 = 0.1393 * m2in; % Mean radii at station 4, in
25
26 %% Calculations
27 H_31 = (A_31 * ft22in2)/(2 * pi * r_m31); % Radial height at station 3.1, in
28 H_4 = (A_4 * ft22in2)/(2 * pi * r_m4); % Radial height at station 4, in
29 r_i31 = r_m31 - H_31/2; % Inner radii of station 3.1, in
30 r_i4 = r_m4 - H_4/2; % Inner radii of station 4, in
31 r_o31 = H_31 + r_i31; % Outer radii of station 3.1, in
32 r_o4 = H_4 + r_i4; % Outer radii of station 3.1, in
33
34 %% Layout
35 AR = 12.073/1.922; % Diffuser area ratio, isentropic table, area ratio at M = 0.05/area ...
    ratio at M = 0.32
36 A_32 = AR * A_31; % Area of station 3.2, ft^2
37
38 % Design B, selected because of fewer fuel nozzles
39 r_o32 = r_o4;
40 r_i32 = sqrt(-((A_32 * ft22in2)/pi) - r_o32^2); % Inner radii of station 3.2, in
41 r_m32 = (r_o32 + r_i32)/2; % Mean radii at station 3.2, in
42 H_32 = r_o32 - r_i32; % Radial height at station 3.2, in
43 N_noz = ((4 * pi^2)/(A_32 * ft22in2)) * r_m32^2; % 16
44
45 %% Diffuser
46 eta_dm = 0.9; % diffusion efficiency
47 Am_A1 = 4;
48 A_m = Am_A1 * A_31; % Area at end of flat wall diffuser, ft^2
49 H_m = 4 * H_31; % Height at end of flat wall diffuser, in.
50 H_r = H_32; % Outer casing height
51
52 y = (H_r - H_m)/2;
53 r_om = r_o32 - y;
54 r_im = r_i32 + y;
55 r_mm = (r_im + r_om)/2;
56
57 eta_d = (eta_dm * AR^2 * (1 - (1/Am_A1)^2) + 2 * (AR * (1/Am_A1) - 1))/(AR^2 - 1);
58 L_d = (AR - Am_A1) * H_31;
59 L_ratio = 9;
60 L_m = (L_ratio * H_31 * r_m31)/r_m32; % in
61 Δ_x = sqrt(L_m^2 - (r_m31 - r_mm)^2);
62 L = Δ_x + L_d;
63
64 %% Air partitioning
65 mdot_mb = 0.1 * kg2lbs; % fuel flow at main burner, lbm/s
66 f_st = 0.0685;
67 mdot_31 = 3.57 * kg2lbs; % mass flow rate at compressor outlet, lbm/s
68 phi_4 = mdot_mb/(f_st * mdot_31);
69
70 T_g = 3200; % Rankine (target liner gas temperature, change to have enough cooling air), ...
    3200 default
71 T_m = 2110; % Rankine
72 T_t4 = 3060; % Rankine
73 T_t31 = 1017; % Rankine
74 T_c = T_t31;
75 Δ_Tmax = (T_t4 - T_t31)/phi_4;
76 phi_sz = (T_g - T_t31)/Δ_Tmax;
77 epsilon_pz = 0.7;
78 phi_pz = phi_sz/epsilon_pz;
79 mu_pz = phi_4/phi_pz;

```



```

80 mu_sz = (phi_4/phi_sz) - (phi_4/phi_pz);
81 mu_total = mu_pz + mu_sz;
82
83 phi = (T_g - T_m)/(T_g - T_c);
84 mu_c = (1/25) * (T_g - T_m)/(T_m - T_c);
85 mu_dz = 1 - (mu_total + mu_c);
86
87 %% Dome and Liner
88 gamma = 1.4;
89 P_t31 = 982.9 * kpa2psia;
90 T_t31 = 1017; % Rankine
91 T_t32 = T_t31;
92 P_31 = P_t31/((1 + ((gamma - 1)/2) * 0.32^2)^(gamma/(gamma - 1)));
93 T_32 = T_t32/(1 + ((gamma - 1)/2) * 0.05^2);
94 T_31 = T_t31/(1 + ((gamma - 1)/2) * 0.32^2);
95 P_32 = P_31 * ((T_32/T_31)^(gamma/(gamma - 1)));
96 P_t32 = P_32 * ((1 + ((gamma - 1)/2) * 0.05^2)^(gamma/(gamma - 1)));
97 P_t4 = 904 * kpa2psia;
98 tplc_mb = (P_t32 - P_t4)/(P_t32 - P_32);
99 atr = mu_sz + mu_dz; % ratio of annulus to reference flow rate
100 alpha_opt = 1 - atr^(2/3) * tplc_mb^(-1/3);
101
102 H_l = alpha_opt * H_r;
103
104 %% Total pressure loss
105 tau_pz = T_g/T_t32;
106 tplc_min = ((mu_pz/alpha_opt)^2) * tau_pz * (2 * tau_pz - 1);
107 margin = (tplc_min/tplc_mb) * 100;
108
109 %% Primary Zone
110
111 U_r = 0.05 * sqrt(gamma * 287 * T_t32/r2k) * m2ft; % reference velocity, ft/s
112 V_j = U_r * sqrt(tplc_mb);
113 r_h = 0.5; % in.
114 C_d = 0.64;
115 alpha_sw = 45; % deg
116 mdot_pz = mdot_31 * mu_pz;
117 r_t = sqrt(r_h^2 + ((mdot_pz - 3 * mdot_mb)/(N_noz * pi * C_d * cosd(alpha_sw))) * ((A_32 ...
    * ft22in2)/mdot_31) * (tplc_mb)^(-1/2));
118
119 if r_t > H_l/2
120     disp("Swirler does not fit within the envelope of the liner.")
121 end
122
123 pitch = (2 * pi * r_m32)/N_noz;
124 S = (2/3) * tand(alpha_sw) * ((1 - (r_h/r_t)^3)/(1 - (r_h/r_t)^2));
125 A_sw = pi * (r_t^2 - r_h^2); % per nozzle
126 L_pz = 2 * S * r_t; % axial length of the primary zone, in.
127
128 %% Secondary Zone
129 qj_qr = tplc_mb;
130 qa_qrs = ((mu_sz + mu_dz)/(1 - alpha_opt))^2;
131 ql_qrs = tau_pz * (mu_pz * alpha_opt)^2;
132 Ymax_djs = 1.15 * sqrt(qj_qr * (1/ql_qrs) * (1 - qa_qrs * (1/qj_qr)));
133 d_js = (1/4) * H_l * (Ymax_djs)^(-1);
134 N_hsz = mu_sz * ((4 * A_32 * ft22in2)/(pi * d_js^2)) * (U_r/V_j);
135 disp(N_hsz/N_noz)
136
137 thetas = asind(sqrt(1 - qa_qrs * (1/qj_qr))); % degrees

```

```

138 d_hs = d_js/(sqrt(C_d) * sind(thetas));           % in.
139 L_sz = 2 * H_l;
140
141 %% Dilution Zone
142 qa_qrd = (mu_dz/(1 - alpha_opt))^2;
143 ql_qrd = tau_pz * ((mu_pz + mu_sz)/alpha_opt)^2;
144 Ymax_djd = 1.15 * sqrt(qj_qr * (1/ql_qrd) * (1 - qa_qrd * (1/qj_qr)));
145 d_jd = (1/3) * H_l * (Ymax_djd)^(-1);
146
147 thetad = asind(sqrt(1 - qa_qrd * (1/qj_qr))); % degrees
148 d_hd = d_jd/(sqrt(C_d) * sind(thetad));           % in.
149 N_hdz = mu_dz * ((4 * A_32 * ft22in2)/(pi * d_jd^2)) * (U_r/V_j);
150 disp(N_hdz/N_noz)
151
152 L_dz = 1.5 * H_l;
153
154 %% Total Length
155 L_l = L_pz + L_sz + L_dz;
156 L_total = L_l + L;   % total axial distance
157
158 %% Plot
159 hold on
160 grid on
161 grid minor
162 ylim([2 8])
163 a_x = [0, Δ_x];
164 a_y = [r_o31, r_om];
165 b_y = [r_i31, r_im];
166 c_x = [Δ_x, Δ_x];
167 c_y = [r_om, r_o32];
168 d_y = [r_im, r_i32];
169 e_x = [Δ_x, L];
170
171 O_y = [r_o32, r_o32];
172 I_y = [r_i32, r_i32];
173
174 spacing = (H_r - H_l)/2;
175 O_y2 = [r_o32 - spacing, r_o32 - spacing];
176 I_y2 = [r_i32 + spacing, r_i32 + spacing];
177
178 pz_x = [L, L + L_pz];
179 sz_x = [L + L_pz, L + L_pz + L_sz];
180 dz_x = [L + L_pz + L_sz, L_total];
181 remainder_x = [L_total, L_total + 1.5];
182 remainder_y = [r_i32, r_o32 - 0.5];
183
184 O_y3 = [r_o32 - spacing, r_o32];
185 I_y3 = [r_i32 + spacing, r_o32 - 0.5];
186
187 s_y = [r_m31, r_mm];
188
189 plot(a_x, a_y, 'k', 'LineWidth', 3)
190 plot(a_x, b_y, 'k', 'LineWidth', 3)
191 plot(c_x, c_y, 'c', 'LineWidth', 3)
192 dump = plot(c_x, d_y, 'c', 'LineWidth', 3);
193 plot(e_x, O_y, 'c', 'LineWidth', 3)
194 plot(e_x, I_y, 'c', 'LineWidth', 3)
195
196 plot(pz_x, O_y, 'k', 'LineWidth', 3)

```

```

197 plot(pz_x, I_y, 'k', 'LineWidth', 3)
198
199 plot(sz_x, O_y, 'k', 'LineWidth', 3)
200 plot(sz_x, I_y, 'k', 'LineWidth', 3)
201
202 plot(dz_x, O_y, 'k', 'LineWidth', 3)
203 plot([L + L_pz + L_sz, L_total + 0.2], I_y, 'k', 'LineWidth', 3)
204
205 pz = plot([L + 0.4, L + L_pz], O_y2, 'b', 'LineWidth', 2); % pz_x
206 plot([L + 0.4, L + L_pz], I_y2, 'b', 'LineWidth', 2)
207
208 sz = plot([L + L_pz + 0.1, L + L_pz + L_sz], O_y2, 'r', 'LineWidth', 2); % sz_x
209 plot([L + L_pz + 0.1, L + L_pz + L_sz], I_y2, 'r', 'LineWidth', 2)
210
211 dz = plot([L + L_pz + L_sz + 0.1, L_total], O_y2, 'g', 'LineWidth', 2); % dz_x
212 plot([L + L_pz + L_sz + 0.1, L_total], I_y2, 'g', 'LineWidth', 2)
213
214 legend('Location','SouthEast');
215 legen2 = legend([dump, pz, sz, dz], {'Dump Diffuser', 'Primary Zone', 'Secondary Zone', ...
    'Dilution Zone'});
216 legen2.FontSize = 10;
217
218 plot(a_x, s_y, 'k', 'LineWidth', 2)
219 plot(remainder_x, O_y, 'k', 'LineWidth', 3)
220 plot([L_total + 0.2, L_total + 1.5], remainder_y, 'k', 'LineWidth', 3)
221
222 plot(remainder_x, O_y3, 'k', 'LineWidth', 2)
223 plot(remainder_x, I_y3, 'k', 'LineWidth', 2)
224
225 ylabel('Radial Length, inches', 'FontSize', 12), xlabel('Axial Length, inches', ...
    'FontSize', 12);

```

B.4 FBG.m

This script evaluates monitoring fibre parameters and plots relevant graphs.

```

1
2 clear
3 clc
4 close all
5
6 % Script written by Jo Wayne Tan to size FBG
7 % of Purpose Built Helicopter. 11/06/2020
8 % FBG Design based on Condition monitoring techniques for fibre mooring ropes
9
10 %% Unit conversions
11 m2ft = 3.28;
12
13 %% Inputs
14 L_r = 25/m2ft; % length of rope
15 LL_f = [0.04:0.01:0.08]; % lay length of optical fibre, 0.06
16 d_f = [0.005:0.0005:0.01]; % helix diameter of optical fibre, 0.08
17
18 %% Plots
19 figure (1)
20 hold on
21 grid on

```

```

22 for i = 1:length(LL_f)
23     alpha = atan(pi .* d_f ./ LL_f(i)); % helix pitch angle
24     epsilon_fr = (cos(alpha)).^2; % axial strain in optical fibre and rope ratio
25     plot(d_f, epsilon_fr, 'LineWidth', 1.5);
26 end
27
28 ylabel('Optic Fibre Strain / Rope Strain', 'FontSize', 12), xlabel('Helix Diameter (m)', ...
    'FontSize', 12);
29
30 figure (2)
31 hold on
32 grid on
33 for i = 1:length(LL_f)
34     L_fr = (1/LL_f(i)) * sqrt((pi .* d_f).^2 + LL_f(i)^2); % length of optical fibre and ...
        rope ratio
35     plot(d_f, L_fr, 'LineWidth', 1.5)
36 end
37
38 ylabel('Optic Fibre Length / Rope Length', 'FontSize', 12), xlabel('Helix Diameter (m)', ...
    'FontSize', 12);
39
40 figure (3)
41 hold on
42 grid on
43 turns = ceil(L_r./LL_f);
44 sloss_curve = 0.05 * turns/100;
45 [AX, H1, H2] = plotyy(turns, LL_f, turns, sloss_curve, 'plot');
46 set(AX(1), 'YLim', [0.04 0.08], 'YTickMode', 'auto')
47 set(AX(2), 'YLim', [0 0.1], 'YTickMode', 'auto')
48 xlim(AX(1), [100 180])
49 xlim(AX(2), [100 180])
50 set(H1, 'LineWidth', 1.5);
51 set(H2, 'LineWidth', 1.5);
52 xlabel('No. of turns', 'FontSize', 12);
53 ylabel(AX(1), 'Lay Length (m)', 'FontSize', 12);
54 ylabel(AX(2), 'Total Attenuation (dB)', 'FontSize', 12);
55
56 %% Calculations
57 d_r = 0.0111125; % rope diameter, m
58 LL_f = 0.06; % lay length of optical fibre, m
59 d_f = 0.008; % helix diameter of optical fibre, m
60 alpha = atan(pi * d_f / LL_f); % helix pitch angle
61 epsilon_fr = (cos(alpha))^2; % axial strain in optical fibre and rope ratio
62 strain_f = 1.6915e-05 * epsilon_fr; % strain in optical fibre
63
64 if strain_f > 0.01
65     disp('Max axial strain exceeded!')
66 end
67
68 L_fr = (1/LL_f) * sqrt((pi * d_f)^2 + LL_f^2); % length of optical fibre and rope ratio
69 rho = (d_f/2)/((sin(alpha))^2); % radius of curvature
70
71 L_f = L_fr * L_r; % length of optical fibre, m
72 sloss_straight = L_f/1000 * 0.3; % dB
73 turns = ceil(L_r/LL_f); % no. of turns
74 sloss_curve = 0.05 * turns/100; % dB, 0.3
75
76 disp(sloss_curve)
77 disp(sloss_straight)

```